

Nova Exponential Decline Model

The **Nova Exponential Decline Model** fits the early decline of a nova outburst with the function used by Kok (2010), equation (10):

$$m(t) = P_1 - P_2 e^{-P_3(t-t_0)}$$

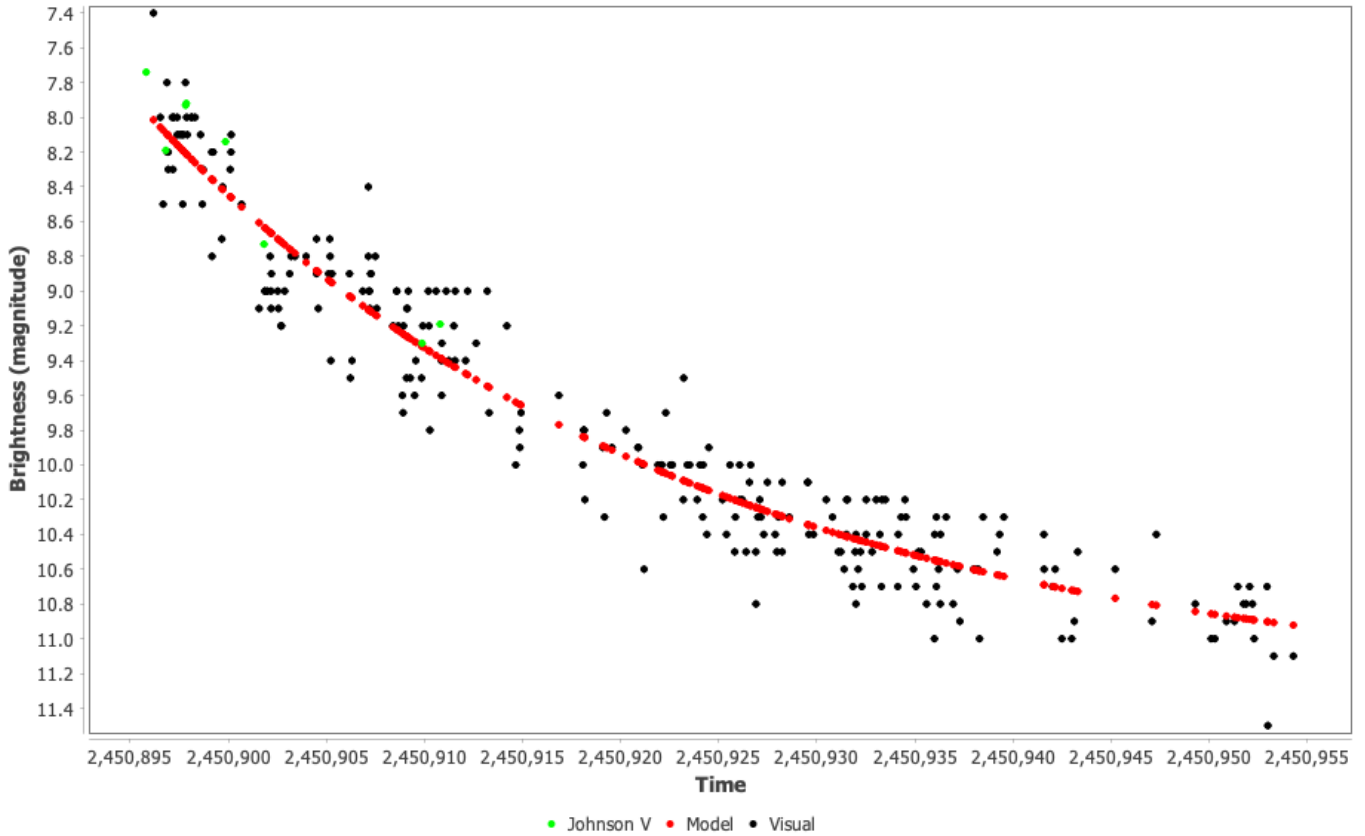
This plug-in appears in VStar's **Analysis** menu as **Nova Exponential Decline Model**. The same model is also used internally by the MMRD nova distance calculator when the **Exponential model fit (Kok 2010, eq. 10)** source is selected.

Parameters

Parameter	Meaning
P_1	asymptotic post-outburst magnitude
P_2	outburst amplitude above the asymptote
P_3	decline rate in inverse days
t_0	fit origin, taken to be the JD of the brightest observation

The model is fitted by non-linear least squares using a Levenberg-Marquardt optimizer. The fit is applied to observations from the brightest observation onward.

Light Curve for V4633 Sgr



The **Analysis > Models** dialog shows the fit as a VeLa function.

Decline Times

The model has a closed-form crossing time. For a decline of Δ magnitudes from reference maximum m_0 :

$$t_{\Delta} = \frac{\ln(P_2/(P_1 - m_0 - \Delta))}{P_3}$$

This is not a separate model: it is equation (10) rearranged. Setting $m(t) = m_0 + \Delta$ and solving for $t - t_0$ inverts the exponential, which is why the forward model (and the VeLa function shown in the **Models** dialog) uses $e^{-P_3(t-t_0)}$ while the crossing time uses a logarithm and a division. t_{Δ} is therefore the elapsed time from the fit origin t_0 .

The MMRD calculator uses this to compute:

- t_2 , where $\Delta = 2$
- t_3 , where $\Delta = 3$

When the model is run directly from the **Analysis** menu, the model series is added to the plot and can be inspected like other VStar models.

Relationship to MMRD Calculator Plug-in

The MMRD nova distance calculator plug-in uses this model to smooth noisy nova decline data before extracting t_2 and t_3 . This is especially useful for raw visual observations from AID, where direct crossing detection can be misled by scatter.

For Kok-style MMRD testing:

1. Load the nova observations over the same JD window used in the paper.
2. Fit the exponential model.
3. Measure t_2 and t_3 from the fitted curve relative to the observed or chart-read maximum.
4. Use those values in the MMRD nova distance calculator.

Uncertainties

The fit produces parameter uncertainties when the optimizer can estimate them. When this model is used as the decline-time source in the MMRD nova distance calculator, those uncertainties are propagated through the closed-form crossing time to populate the calculator's σt_2 and σt_3 fields. The full error-bar pipeline (decline times to absolute magnitude to distance bounds) is documented with the MMRD nova distance calculator plug-in.

The parameter uncertainties are used for that propagation only; they are not shown in the **Models** dialog.

Fit Metrics

Like other VStar models, the fit reports three standard goodness-of-fit metrics, computed for the three estimated parameters P_1 , P_2 , P_3 :

- **RMS** — the root-mean-square of the residuals about the fitted curve. This is a direct measure of how closely the model follows the observed decline; a large RMS relative to the photometric scatter suggests the exponential form or the chosen JD window is a poor fit.
- **AIC** and **BIC** — the Akaike and Bayesian Information Criteria. These are intended for *comparing competing fits of the same data* (for example this model against another model series). Their absolute values are not meaningful in isolation; lower values indicate a better complexity-penalised fit when comparing alternatives.

These metrics describe overall fit quality and are distinct from the parameter covariance used above: RMS/AIC/BIC summarise how well the curve matches the data, whereas the covariance is what propagates into the σt_2 and σt_3 (and ultimately the MMRD distance) uncertainties.

When to Use This Model

Use this model when:

- the nova has a reasonably smooth decline after maximum,
- you need t_2 and t_3 for MMRD distance estimation,
- raw observations are too noisy for direct peak+2/peak+3 crossing detection,
- you want a visible model series for inspection before running the MMRD tool.

Limitations

- The model is empirical and intended for the early decline phase.
- It requires enough post-maximum observations to constrain three parameters.
- If the fitted amplitude does not reach peak + 2 or peak + 3 magnitudes, the corresponding t_2 or t_3 is unavailable.
- The fit origin is the brightest observation, so a spurious bright point can affect the model. Review the input data and filters before fitting.

Reference

Kok, Y. 2010, *Absolute Magnitudes and Distances of Recent Novae*, JAAVSO, 38, 193.